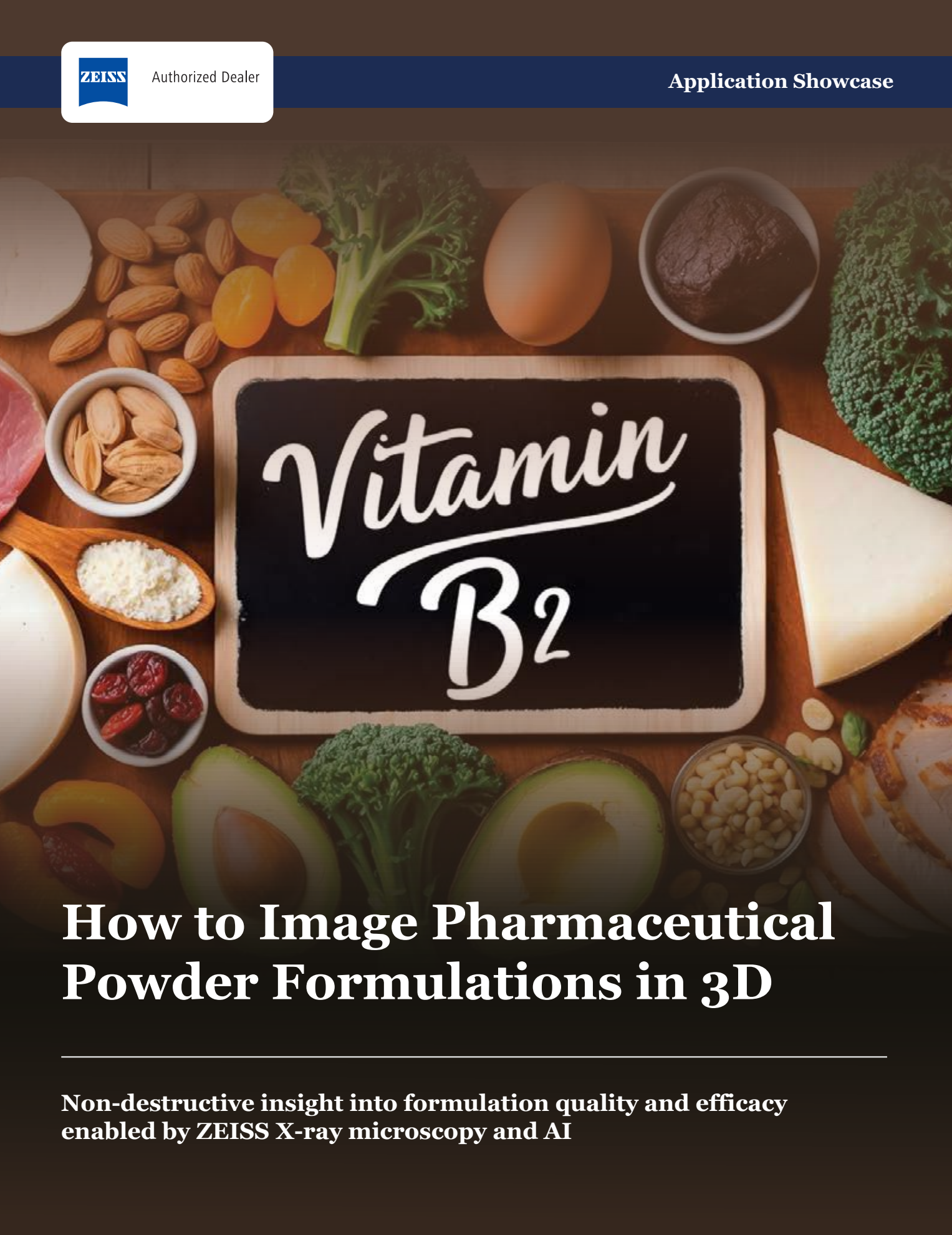




Vitamin
B₂

How to Image Pharmaceutical Powder Formulations in 3D

Non-destructive insight into formulation quality and efficacy
enabled by ZEISS X-ray microscopy and AI

Dr Ria L. Mitchell, Andy Holwell

ZEISS Research Microscopy Solutions, Cambourne, UK

Consistent quality and efficacy are essential in drug formulations for human health. Therefore, easy-to-use, fast, and reliable quantitative characterization methods are needed to ensure pharmaceutical formulations contain the correct materials in the correct doses. This could include ensuring the correct particle sizes, verifying the proportion of active materials and excipients, and their production to specification. Here, a novel 3D imaging and phase identification-based workflow is introduced. 3D X-ray Microscopy (XRM) is used to non-destructively image riboflavin (vitamin B2) capsule powders at high resolution. Following this, the 3D data is reconstructed using two novel AI-based methods for de-noising, contrast improvement, phase enhancement, and phase artefact removal.

Finally, a phase classification, segmentation, and quantification method is used to understand the formulations in 3D. Further, Scanning Electron Microscopy (SEM) and Energy Dispersive X-ray Spectroscopy (SEM-EDS) are used as a ground truth and to provide additional chemical information. These approaches form a unique workflow for the deeper understanding of pharmaceutical formulations and to ensure efficacy and QA/QC in the industry.

Keywords or phrases:

3D X-ray Microscopy (XRM), non-destructive pharmaceutical characterization, AI-driven image reconstruction and segmentation, pharmaceutical powder formulation analysis, quality assurance and efficacy assessment, phase identification and particle morphology

INTRODUCTION

1. Pharmaceutical powder formulations and riboflavin background

Pharmaceutical powders are dry particulate formulation mixtures of active pharmaceutical ingredients (APIs) and

excipients. Powders can refer to an individual type of dosage form (e.g., orally administered powders such as inhalation/nasal delivery blends, powders within capsules) or the raw materials that are later manufactured into other dosage forms (e.g., pressed tablets). Powder characterization at an early stage of the production process is important to understand the structure of the raw materials that will eventually become manufactured products, through to ensuring quality assurance and quality control (QA/QC) of the finished product for efficacy reasons.

For example, while APIs generally have a smaller particle size compared with excipients¹, API and excipient powder particle size, shape and surface features influence flow during production², and at the other end of the pipeline, a large particle size distribution will affect dissolution and ultimately how well the drug is taken up by the body³. Additionally, the quantities and balance of various excipients is important. Therefore, the right characterization methods should be used before, during, and after the powder production process. Here, an example of 3D capsule powder characterization within a riboflavin (vitamin B2) migraine relief supplement is shown.

Riboflavin naturally occurs in food groups such as dairy products, meat, and nuts. It is a water-soluble micronutrient that is essential for the reproduction of cells and growth, and the prevention of inflammation, cancer, and sepsis, whilst also acting as an antioxidant^{4,5}. It has also been found to contribute towards the alleviation of migraine symptoms⁴ by the reduction of oxidative stress, mitochondrial dysfunction, and neuroinflammation⁴⁻⁸, and is therefore being increasingly administered to patients because of its preventative effects⁴.

While it is common in multi-vitamins and dietary supplements, numerous manufacturers now produce riboflavin specifically as a migraine relief product, which is facilitated by a cost-effective single fermentation production step⁹. As well as the active riboflavin particles, variable excipients are also present; these include anti-caking (or flow) agents, designed to act as lubricants and facilitate a homogeneous mix, and binders, fillers, or diluents, which add cohesivity. Because of the various ways in which

riboflavin is manufactured between manufacturers, and the variety of excipient additives incorporated, characterization and quantification is needed to understand their structure, distribution, and effectiveness¹⁰.

In this work, the aim is to quantify and characterize powders using a simple 3D workflow to understand particle structure, morphology, and distribution. Two capsules from two different manufacturers are used; capsule 1, which reportedly was ineffective, and capsule 2, of good-reported efficacy. Firstly, X-ray Microscopy (XRM) is undertaken to image the powders non-destructively in 3D. Secondly, various advanced deep learning and AI based reconstruction approaches, namely ZEISS DeepRecon Pro and ZEISS PhaseEvolve, are applied to the 3D data to enhance data quality. Finally, an automated phase classification method (ZEISS Phase Identifier 3D) is applied to understand the chemical and phase distributions of the different components in 3D. This approach highlights the benefits of using 3D imaging, deep-learning and AI to enhance 3D data, and automated phase classification for segmentation and quantification of pharmaceuticals. This workflow lends itself to powder, capsule, and pressed tablet characterization.

2. Materials and methods

The first step of the workflow is the collection of a series of non-destructive multi-scale X-ray Microscopy (XRM) scans using a ZEISS VersaXRM (summary in Figure 1). Powders were decanted into polyimide tubing for ease of scanning and thermal stability and were packed to prevent static movement during scanning. Two samples were scanned from two different manufacturers: capsule 1 and capsule 2, both of which reportedly contained riboflavin (API), and Mg stearate (likely as a lubricant). At this point, there was no additional information regarding how each of the powders were manufactured.

Multi-scale scans were taken at 4.23, 1.50, and 0.72 μm voxel sizes, where 801 projections were collected per scan at bin 2 (Figure 2). Further higher resolution tomographies were obtained for capsule 2; these were collected using the 20 $\times$ objective at 607 nm voxel size.

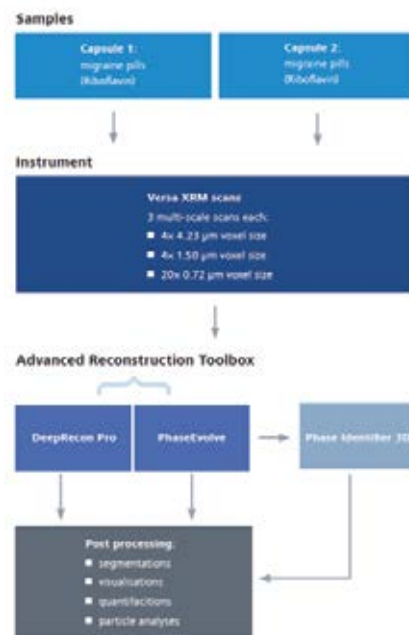


Figure 1: Summary of the workflow used in this study.

Initial qualitative investigation of the XRM data reveals that each set of powders appear to have very different morphological characteristics; capsule 1's powders being crystalline, porous and angular, and capsule 2's powders having a larger range of sizes, non-porous and rounded (Figure 2). This difference may be the result of different manufacturing processes, capsule 1 likely being formed by spray drying, and capsule 2 likely being formed by granulation.

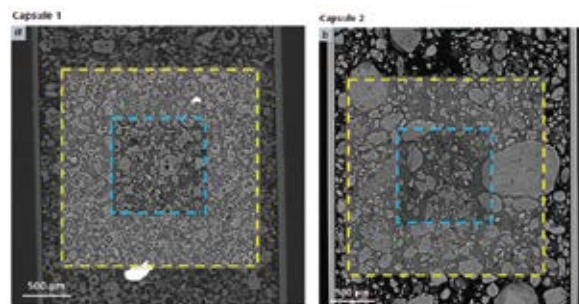


Figure 2: 2D slice view of the position of the three successively higher resolution multi-scale scans. Low resolution, full field of view scan is 4.23 μm voxel size, yellow outlined scan is 1.50 μm , and blue outlined high-resolution scan is 720 nm.

Further 2D imaging and analysis was undertaken using a ZEISS Sigma 560 VP field emission scanning electron microscope (SEM), equipped with an Oxford Instruments Ultim Max 170 EDS detector.

3. AI and deep-learning reconstruction for de-noising, contrast improvement, and low-z phase enhancement

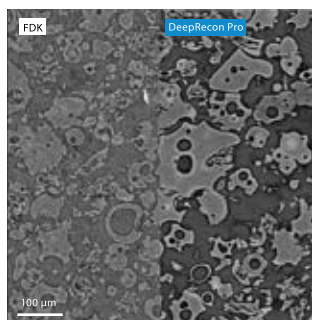


Figure 3: 2D slice comparison of FDK and ZEISS DeepRecon Pro reconstructed data for capsule 1. This high-resolution dataset was collected at 720 nm voxel size. Adapted from [10].

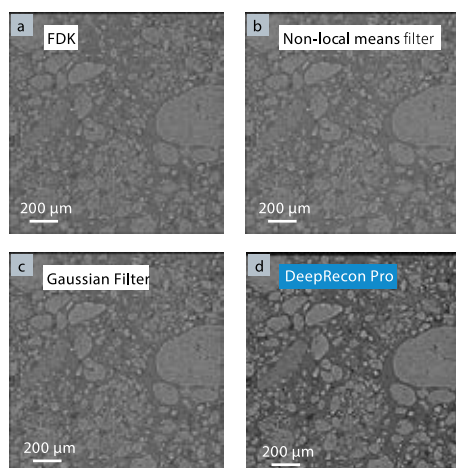


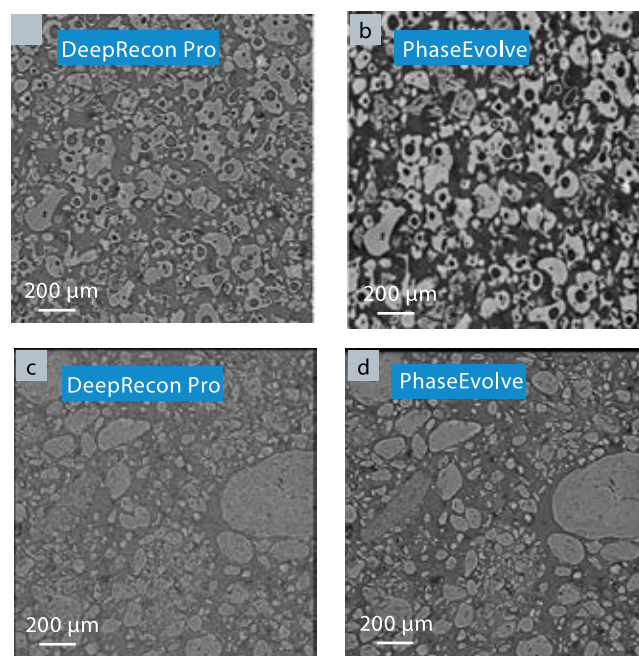
Figure 4: Comparison of FDK (a), two generic post-processing filter types (Nonlocal means and gaussian) (b, c), and the DeepRecon Pro reconstruction (d) results for capsule 2 powders.

Additionally, ZEISS PhaseEvolve is used to complement ZEISS DeepRecon Pro reconstructed data to a) improve contrast between low-density particles, b) remove unwanted phase fringe artefacts (“Fresnel Fringes”) from the outside of particles, and c) reveal differences in absorption at low kV which can be related to subtle chemical compositional differences^{12, 13}.

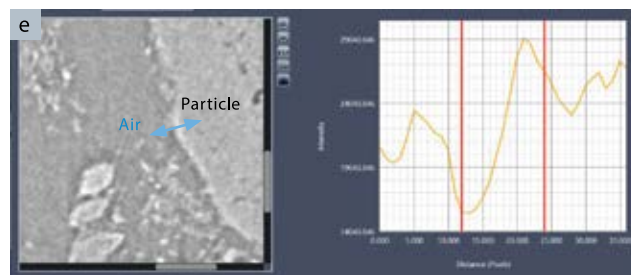
It is particularly important to remove the unwanted phase fringe artefacts which can contribute towards erroneous segmentations downstream. It works by applying an AI

algorithm to data to enhance the contrast enough that it exposes different material contrasts uniquely revealed by XRM and removes phase effects using an analytical description of the associated artefact¹². In the data presented here, it not only removes the unwanted phase fringe artefacts, but also reveals an additional particulate phase in capsule 2 (Figure 5, 6) and leads to more accurate segmentations of the particulate phase compared with FDK (Figure 6).

Also shown is how PhaseEvolve smooths greyscale fluctuations and noise at particle boundaries to eliminate the Fresnel Fringes (Figure 5e-f). DeepRecon Pro and PhaseEvolve are complementary, resulting in more accurate data for segmentation and quantification.



FDK Reconstruction



PhaseEvolve Reconstruction

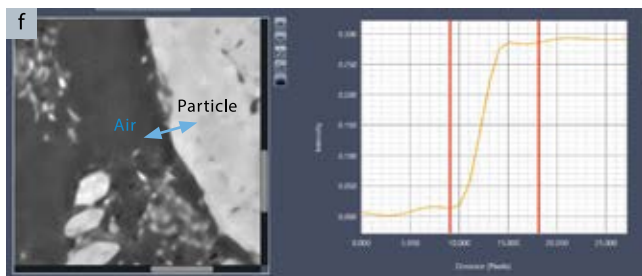


Figure 5: 2D slice comparisons of DeepRecon Pro and PhaseEvolve for capsule 1 (a, b) and capsule 2 (c, d). Phase fringes are eliminated, and a new phase previously unseen is revealed for capsule 2 using PhaseEvolve (d). Also shown are line plots of grayscale variations in FDK (e) and PhaseEvolve (f) reconstructed data across a particle – air boundary.

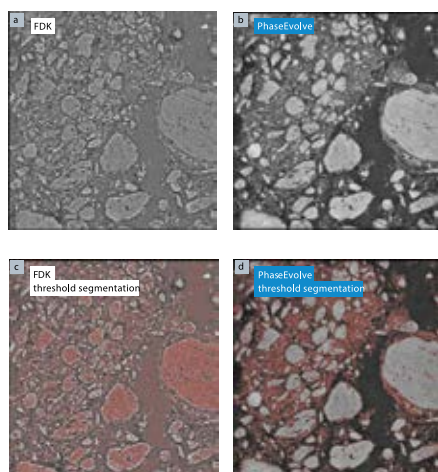


Figure 6: Comparison of (a) FDK, (b) PhaseEvolve, (c) FDK threshold segmentation, and (d) PhaseEvolve threshold segmentation in a high-resolution scan of capsule 2. Data collected using the 20× objective at 607 nm voxel size. Images show the segmentation of needle shaped particulates is more successful in the PhaseEvolve enhanced data.

4. 3D phase classification, segmentation and quantification of pharmaceutical materials

ZEISS Phase Identifier 3D software allows for the automated phase identification, classification, segmentation, and quantification of pharmaceutical materials, as well as others^{14–16}. It works by first utilizing DeepRecon Pro to enhance contrast-to-noise for phase discrimination. Following the creation of a bespoke phase library, the software then uses AI-based methods to identify the unique phase locations on peaks on the data histogram, based on

linear attenuation values¹⁵ (Figure 7). These phases are then segmented and quantified, and accurate phase-based particle analysis can be completed.

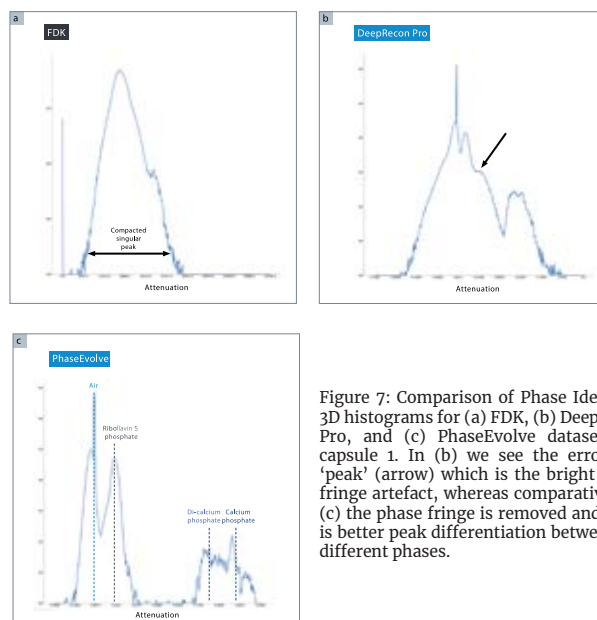


Figure 7: Comparison of Phase Identifier 3D histograms for (a) FDK, (b) DeepRecon Pro, and (c) PhaseEvolve datasets for capsule 1. In (b) we see the erroneous 'peak' (arrow) which is the bright phase fringe artefact, whereas comparatively in (c) the phase fringe is removed and there is better peak differentiation between the different phases.

The PhaseEvolve dataset was used in Phase Identifier 3D for segmentation of capsule 1 because it clearly identifies riboflavin as the lowest density phase (besides air; Figure 7) but also reveals three other findings. Firstly, nothing on the ingredients list matched with the highest density/attenuating phase (i.e., the peak furthest to the right), suggesting something is in the sample that wasn't listed on the ingredients. Subsequent SEM imaging and SEM-EDS analysis was undertaken to help identify the higher density phase (Figure 8a–c), which was identified as calcium phosphate, commonly used as a filler. This phase was then added to the Phase Identifier 3D phase library for classification and resulted in a peak match (Figure 7c). So, the combination of SEM and SEM-EDS analysis, and Phase Identifier 3D classification, identified the presence of a phase which is not listed on the ingredients in this specific supplement. Secondly, segmentation of the DeepRecon Pro data revealed that the minor peak (arrow in Figure 7b) was in fact the phase (Fresnel) fringe artefact surrounding the particles, and not 'real' data at all. In the enhanced PhaseEvolve dataset, this minor peak has been eliminated, leaving only the correct phase retrievable data for

segmentation, highlighting the need to use artefact free data. Finally, magnesium stearate, which was listed on the ingredients, is not identifiable from Phase Identifier 3D, indicating it is not present in at least this capsule (or it could be present but in proportions beyond the resolution of this dataset). Magnesium stearate is generally used as a lubricant for flow and/or as an anti-caking agent, which could indicate why the capsules were ineffective (i.e., limited flowability, lack of homogenisation). These results exhibit why it is important to enhance pharmaceutical data using PhaseEvolve before segmentation and quantification.

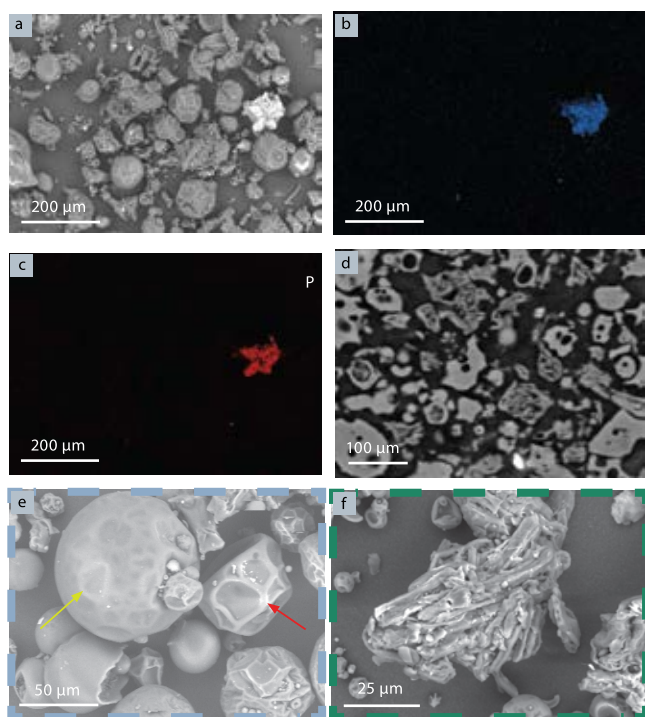


Figure 8: Complimentary SEM imaging and SEM-EDS analysis to establish the composition of the unknown, high-density phase in capsule 1 (yellow circle in (a)). (a) SEM image of capsule 1 powder, (b, c) SEM-EDS reveals the composition of this high-density grain as calcium phosphate, (d) shows a 2D slice through the XRM data highlighting the two different particle types, flakey (green) and rounded (blue), and (e, f) show SEM comparative ground truth images of the particles to validate the presence of the variable particle types. Arrows in e: yellow = 'intact' particle, red = collapsed particle examples.

5. Particle analysis of pharmaceutical powders following AI-based segmentation

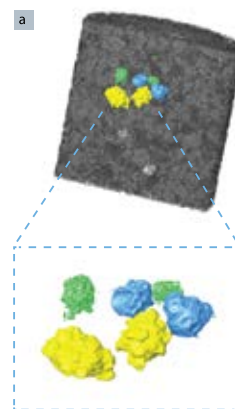
The XRM data reveals that while some of the powders in capsule 1 are intact and are near spherical, some powders are

collapsed with shrinkage (Figure 8, 9). This variation is likely due to different drying rates during the manufacturing process, likely at a lower-than-normal temperature for the collapsed powders, and variation in the characteristics of the powder crusts (skin) 200 μm 200 μm P 200 μm Ca 100 μm 50 μm 25 μm a b c d e f¹⁷ leading to a range of morphologies and rates of collapse.

Many of the particles in capsule 1 are also porous and have hollow interiors; this increases surface area and may beneficially allow for greater solubility and dissolution. This porosity would not be identifiable from using SEM alone, and benefits from a non-destructive 3D XRM analysis.

Direct comparison of the segmented particle characteristics in the two capsules (Figure 10) highlights the morphological differences resulting from the different production methods employed by each manufacturer. This highlights that there can be considerable variation in the powders used in vitamin supplements, which ultimately can affect how effective the APIs are.

	Capsule 1	Capsule 2
Voxel Size	1.50 μm	1.50 μm
Field of View	1.5 x 1.5 mm	1.5 x 1.5 mm
Spatial resolution (4x objective)	1.9 μm	1.9 μm
Total Active Particles (Riboflavin)	3,40,647	2,04,229
Total Particles >4.5 μm volume	77,247	1,60,671
mean particle volume	164,301 μm^3	6672 μm^3
range particle volume	10,116,343 μm^3	52390178 μm^3
Average particle aspect ratio	0.43	0.36
Average feret diameter	20.42	12.78



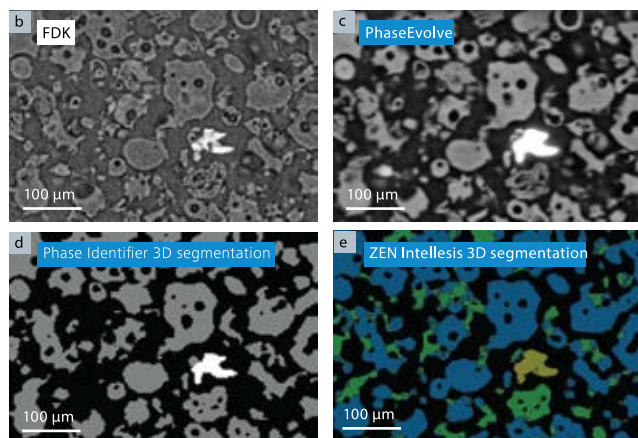


Figure 9: Example of how a combination of AI-based processes (PhaseEvolve, Phase Identifier 3D, and ZEN Intellesis) in a correlative arrangement can provide accurate segmentations of the different particle types in pharmaceutical powders. Blue = rounded particles, green = flaky particles, yellow = calcium phosphate. Width of a) is 2 mm.

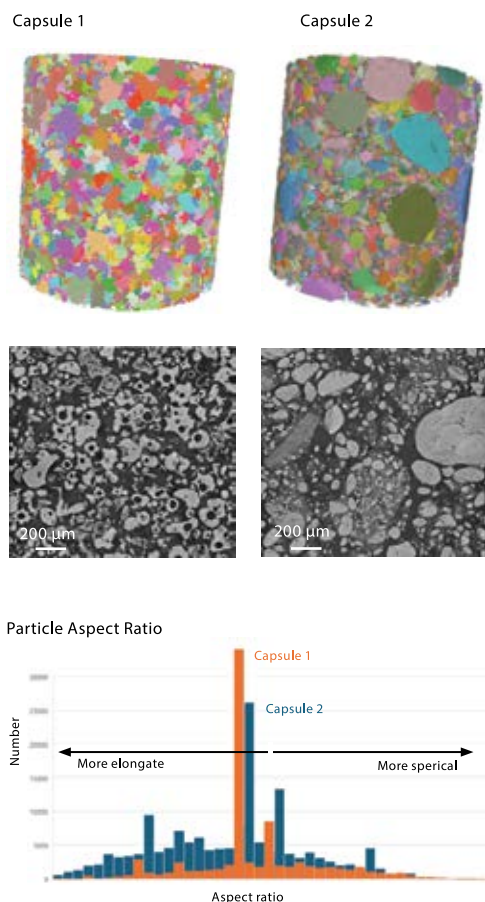


Figure 10: Particle analysis comparison of capsules 1 and 2, highlighting the different particle sizes, structures, and shapes between the two different manufacturers. Only particles 3× the voxel size ($>4.5 \mu\text{m}$) included.

Phases from Phase Identifier 3D segmentation	Contents (%)
Riboflavin	99.16
Ca Phosphate	0.84

Phases from further Intellesis segmentation	Contents (%)
Riboflavin flakey (less Na)	26.43
Riboflavin rounded (more Na)	72.95
Ca Phosphate	0.62

Table 1: 3D composition of capsule 1 based on Phase Identifier 3D and Intellesis segmentations.

SUMMARY

Direct comparison of the segmented particle characteristics in the two capsules (Figure 10) highlights the morphological differences resulting from the different production methods employed by each manufacturer. This highlights that there can be considerable variation in the powders used in vitamin supplements, which ultimately can affect how effective the APIs are.

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